

PREDICT ACADEMIC SUCCESS AND DROP OUT

PROBLEM STATEMENT

- **PREDICTING STUDENT DROPOUT:**
An Early Warning System for Higher Education
- **OUR SOLUTION:**
A machine learning model that identifies at-risk students at enrollment & after 1st semester
- **Impact of the Model:**
Early intervention
Targeted resources
Improved outcomes



DATASET OVERVIEW

DATASET: Student Dropout & Academic Success

Source: UCI Machine Learning Repository (2021)

Dataset Overview

- 4424 students
- 36 features (78 after)
- Mixed variables
- No missing values

TARGET

Three outcomes:

- Dropout
- Enrolled
- Graduate

FEATURE CATEGORIES

Demographics	Academic	Socioeconomic
AGE	Prev grades	Parents' Education
Gender	Course type	Parents' Qualification
Marital Status	Performance	Financial Debt
Nationality	Admission	Scholarships

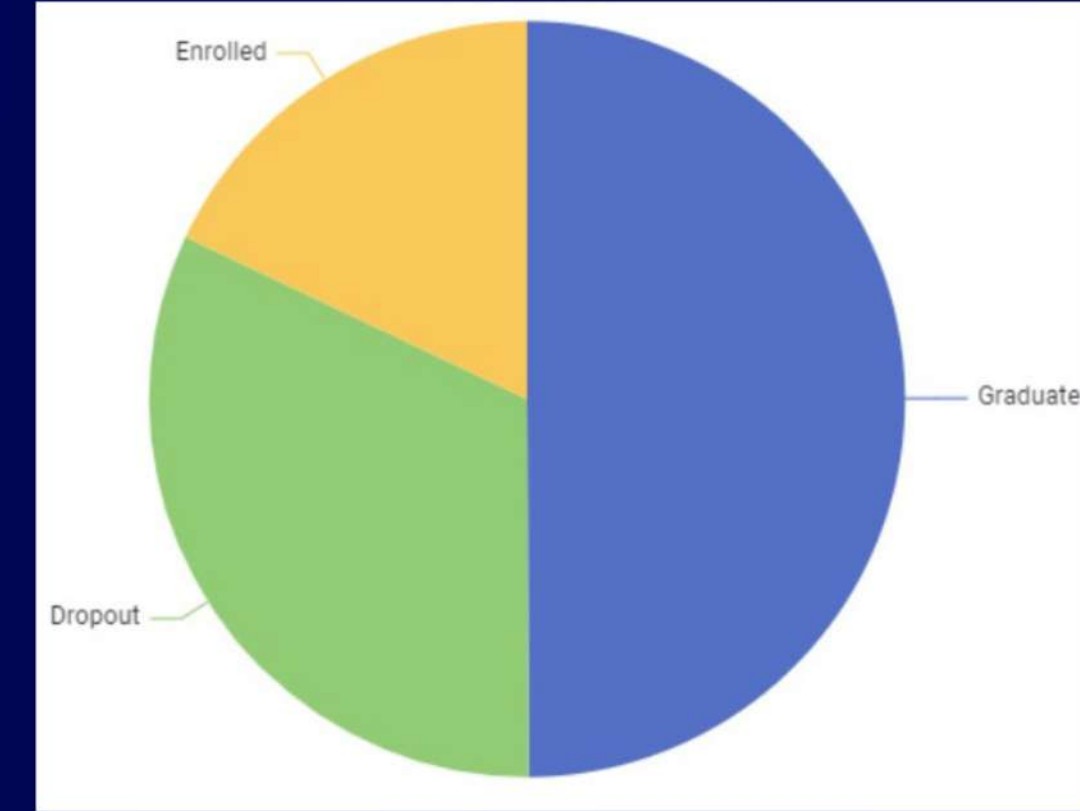
KEY CHALLENGE: Class imbalance & categorical encoding (numerical codes masking categories)

KEY INSIGHTS FROM EDA

WHAT THE DATA REVEALED?

Class Distribution of Target Variables

- Enrolled - 18%
- Dropout - 32%
- Graduate - 50%



FEATURE ENGINEERING:

- *Decoded 7 categorical variables (education, occupation, course, application type)*
- *78 meaningful features after encoding*
- *Preserved interpretability for action*



ML SOLUTION & RESULT

Our Approach

- Built an end-to-end pipeline in KNIME
- Preprocessing: missing values → encoding → normalization → SMOTE balancing
- Trained 3 models: Decision Tree, Random Forest, Gradient Boosting

Performance Results

Model	Accuracy	F - measure	Precision	Recall
Decision Tree	67%	• Dropout - 69% • Graduate - 79% • Enrollment - 41%	• Dropout - 69% • Graduate - 78% • Enrollment - 41%	• Dropout - 65% • Graduate - 80% • Enrollment - 41%
Random Forest	77%	• Dropout - 79% • Graduate - 85% • Enrollment - 45%	• Dropout - 81% • Graduate - 76% • Enrollment - 71%	• Dropout - 77% • Graduate - 96% • Enrollment - 33%
Gradient Boost	77%	• Dropout - 78% • Graduate - 85% • Enrollment - 52%	• Dropout - 80% • Graduate - 78% • Enrollment - 67%	• Dropout - 75% • Graduate - 93% • Enrollment - 43%
Logistic Regression	66%	• Dropout - 67% • Graduate - 80% • Enrollment - 0%	• Dropout - 53% • Graduate - 80% • Enrollment - 0%	• Dropout - 89% • Graduate - 81% • Enrollment - 0%

Best Model: Gradient Boost (77% Accuracy)

- Captures 81% of actual dropouts
- Minimizes false negatives (fewer at-risk students missed)
- Strong balance across all student categories

Confusion Matrix of Gradient Boost Model

	Dropout	Graduate	Enrollment
Dropout	247	31	0
Graduate	82	338	0
Enrollment	134	53	0

CONCLUSION & NEXT STEPS

WHAT WE ACHIEVED:

- 77% accurate dropout prediction model
- Identifies 81% of at-risk students early
- Interpretable features for intervention

KEY RECOMMENDATIONS:

AT ENROLLMENT, Flag students with:

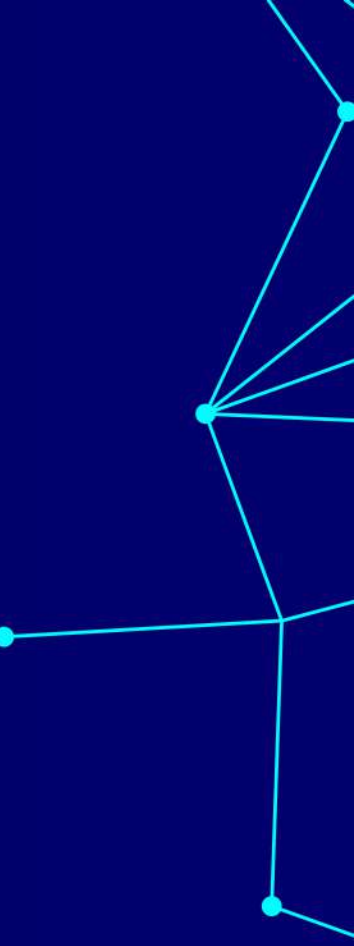
- low parent education,
- Older age (>23), financial issues
- Proactive: Assign mentor, financial counsel

AFTER 1ST SEMESTER:

- Critical checkpoint! Run model with grades
- Students <50% pass rate → intensive support
- Tutoring, study skills workshops, counseling

FUTURE IMPROVEMENTS :

- Real-time dashboard for advisors
- Incorporate mid-semester check-ins
- Test across different institutions
- Personalized intervention recommendations



THANK YOU
